Application-Specific Computer Architecture

(ELL782,ELL7282)

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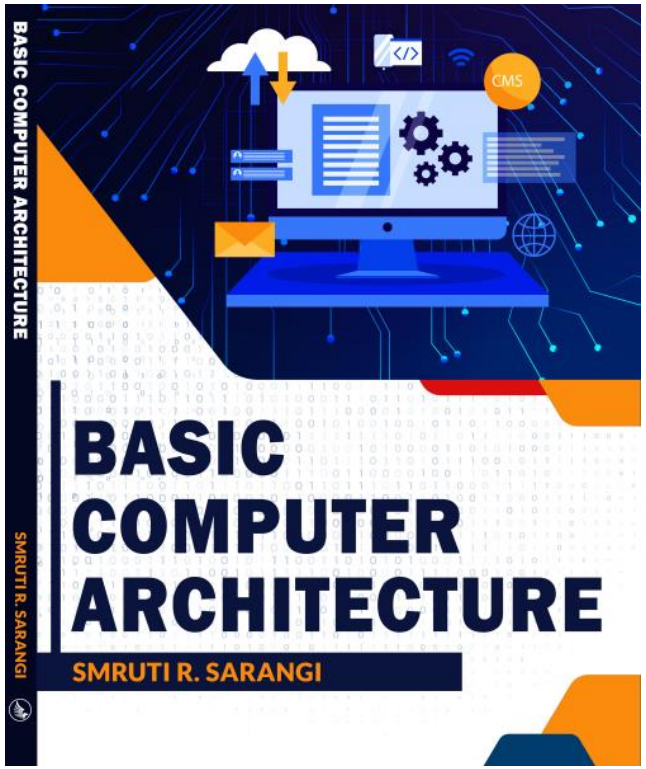
Logistics

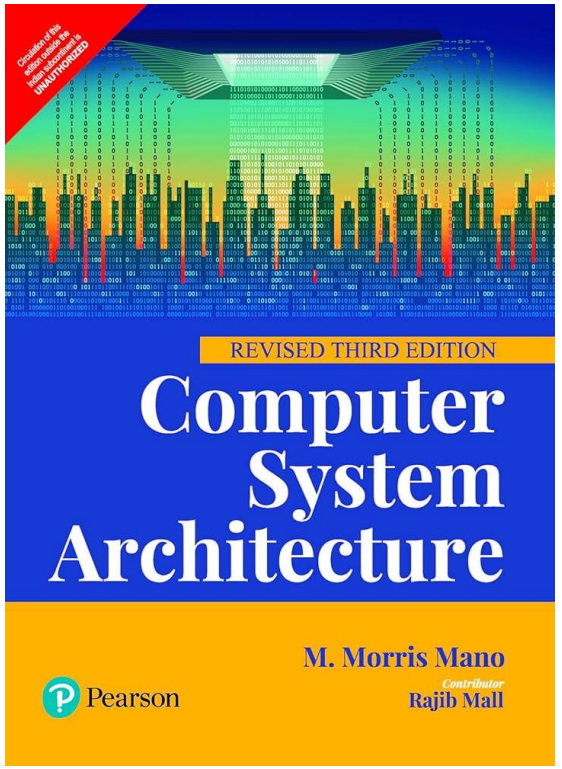
Regular lecture slot: Mon, Thu, 8-9:30 am
Venue: II 336
TAs: Raj Singh Tomar, Indraayudh Talunkar

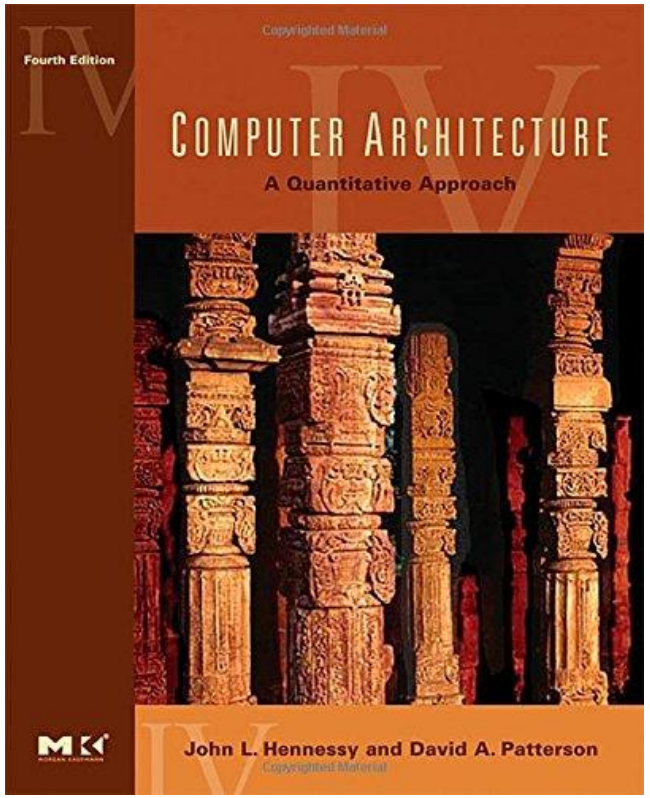
Course website: https://lcs2-iitd.github.io/ELL7282-2601/

Piazza Link: https://piazza.com/iit_delhi/fall2026/ell7282ell782
(Code: carch2601)

Textbooks







Slides are primarily adopted from this book

Tentative Syllabus

Introduction
Language of Bits
Assembly Languages
Digital Logic
Computer Arithmetic
Processor Design
Pipelining
Memory System
Multiprocessor Systems *
I/O and Storage Devices *

*If time permits

Tentative Grading Scheme

- Assignment 1 (10 marks) – Before Midsem
- Assignment 2 (10 marks) – After Midsem
- Four Quizzes (20 marks) – Two before Midsem, two after Midsem
- Daily Quiz: 5 marks
- Attendance: 5 marks
- Midsem: 25 marks)
- Endsem: 25 marks)

- Attendance: 75% (to be eligible to appear in the exam)

What is Computer Architecture ?

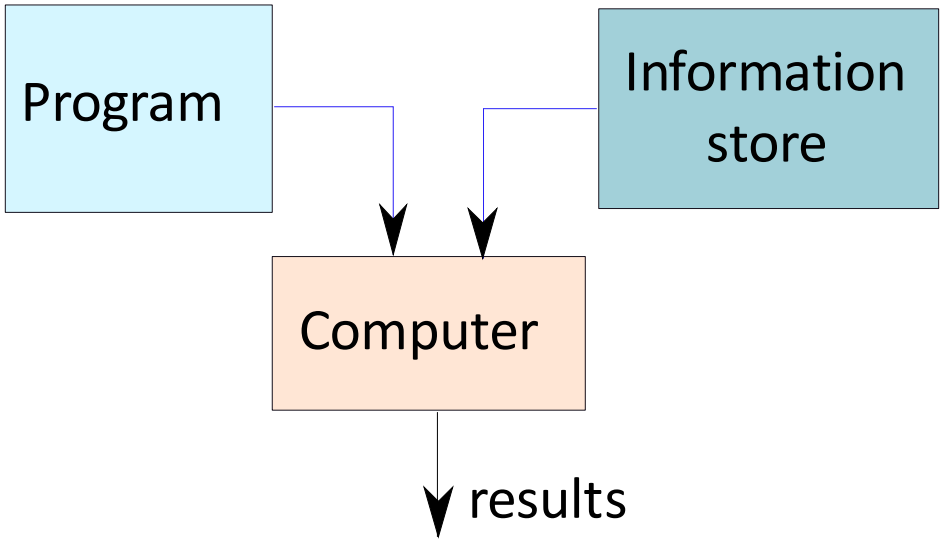
Answer : It is the study of computers ?

* Computer Architecture
  * The view of a computer as presented to software designers
* Computer Organization
  * The actual implementation of a computer in hardware.

What is a Computer ?

A computer is a general purpose device that can be **programmed** to **process** information, and yield **meaningful** results.

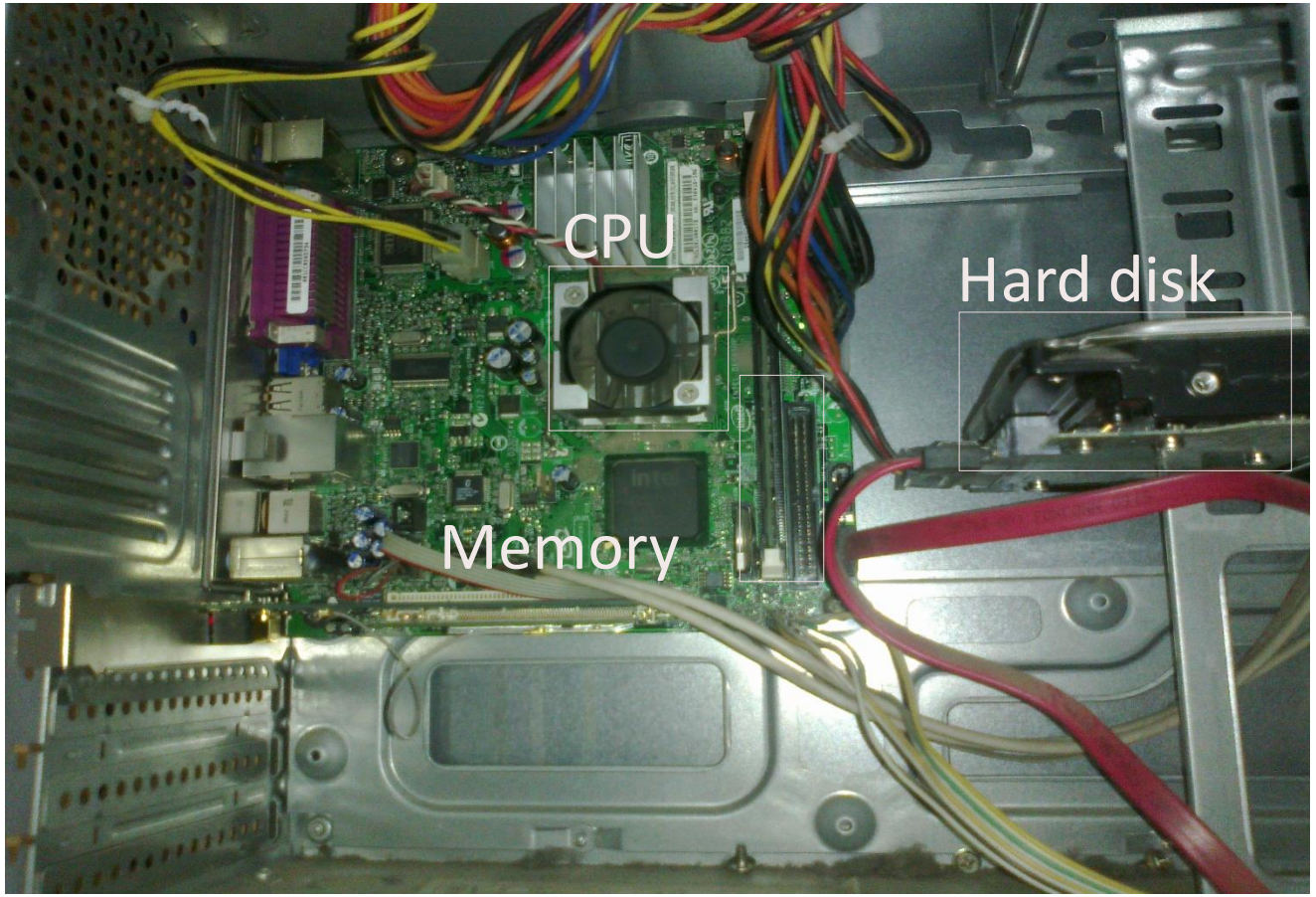
How does it work ?



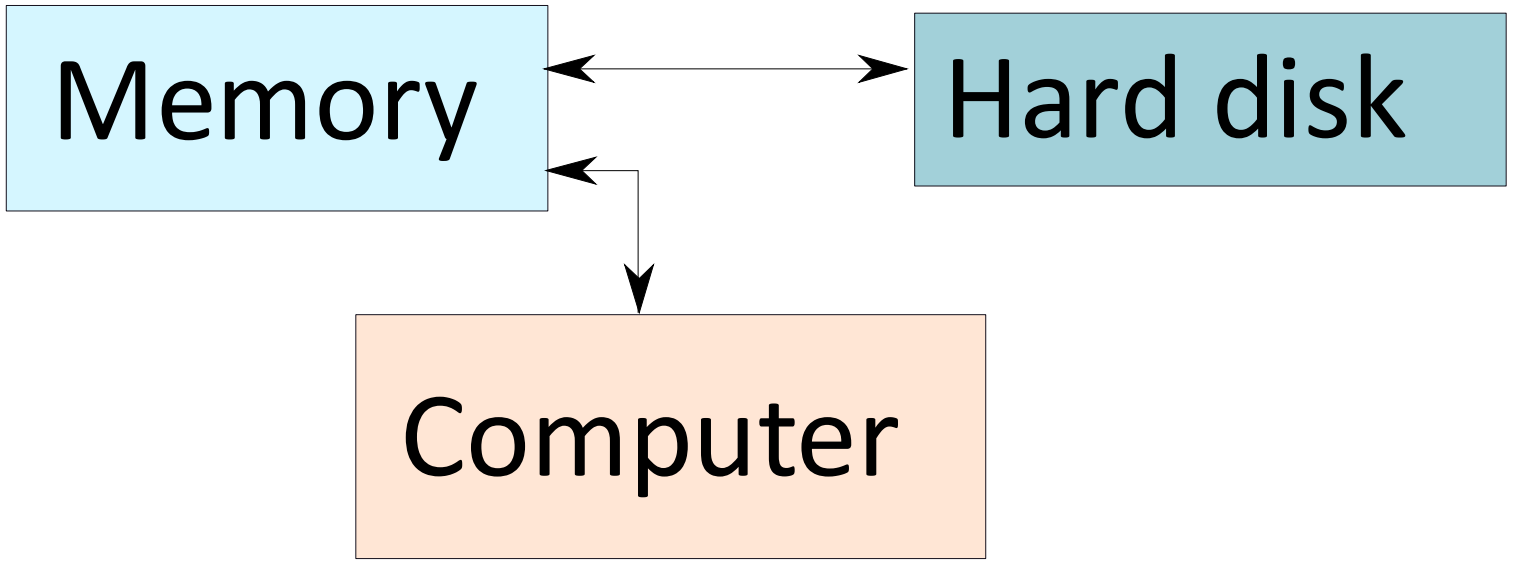
* Program – List of instructions given to the computer
* Information store – data, images, files, videos
* Computer – Process the information store according to the instructions in the program

What does a computer look like ?

* Let us take the lid off a desktop computer

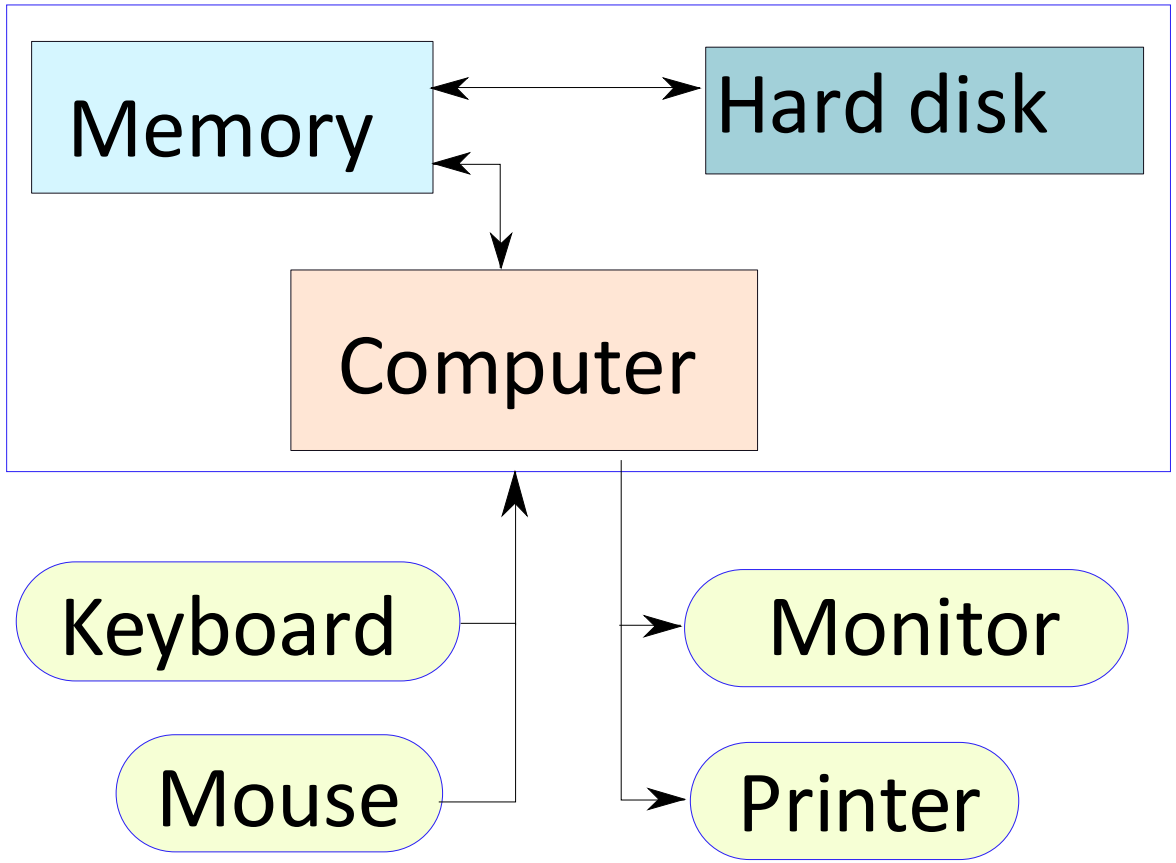


- FPU (Floating Point Unit)
- ISP (Image Signal Processor)
- Microcontrollers (MCU)
- Network Processor / NIC Processor
- Storage Controllers

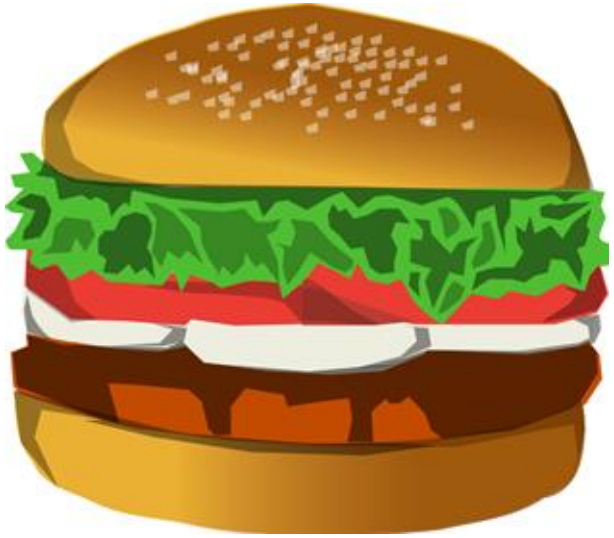


* Memory – Stores programs and data. Gets destroyed when the computer is powered off
* Hard disk – stores programs/data permanently

Let us make it a full system …

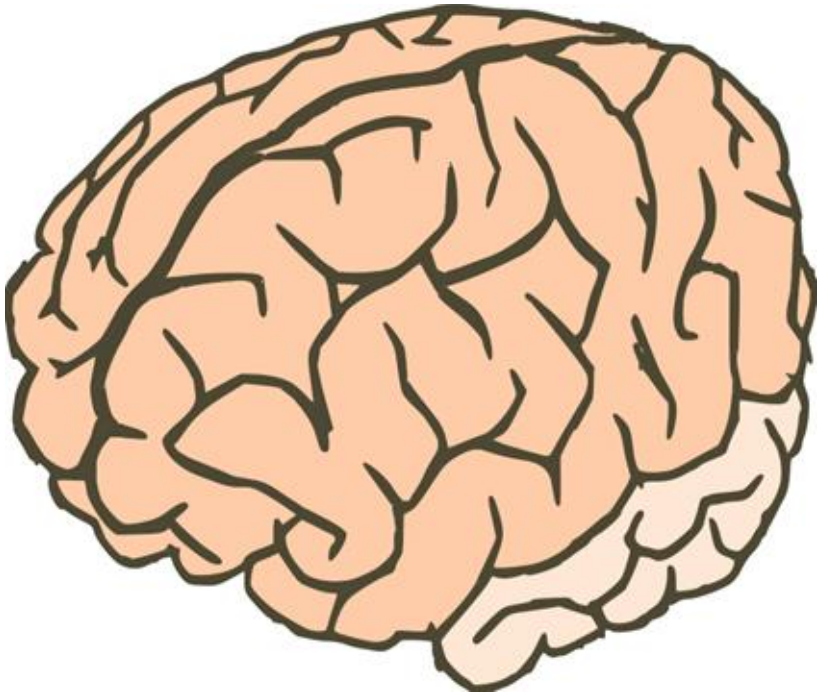


Food for Thought…



* What is the most intelligent computer ?

Answer …



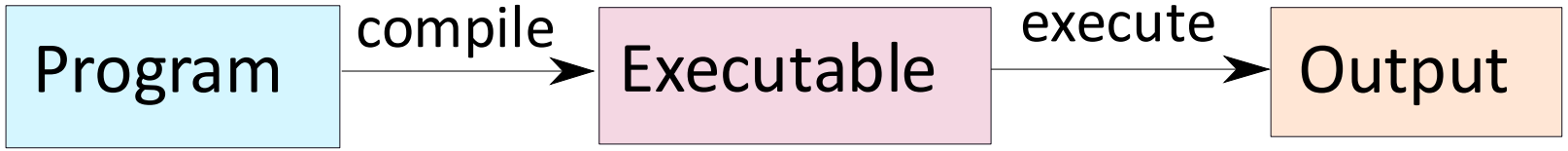
* Our brilliant brains

How does an Electronic Computer Differ from our Brain ?

Feature	Computer	Our Brilliant Brain
Intelligence	Dumb	Intelligent
Speed of basic calculations	Ultra-fast	Slow
Can get tired	Never	After sometime
Can get bored	Never	Almost always

* Computers are ultra-fast and ultra-dumb

How to Instruct a Computer ?



* Write a program in a high level language – C, C++, Java
* **Compile** it into a format that the computer understands
* Execute the program

What Can a Computer Understand ?

* Computer can clearly NOT understand instructions of the form
  * Multiply two matrices
  * Compute the determinant of a matrix
  * Find the shortest path between Mumbai and Delhi
* They understand :
  * Add *a + b* to get *c*
  * Multiply *a * b* to get *c*

The Language of Instructions

* Humans can understand
  * Complicated sentences
    * English, French, Spanish
* Computers can understand
  * Very simple instructions

The semantics of all the instructions supported by a processor is known as its instruction set architecture (ISA). This includes the semantics of the instructions themselves, along with their operands, and interfaces with peripheral devices.

Features of an ISA

* Example of instructions in an ISA
  * Arithmetic instructions : add, sub, mul, div
  * Logical instructions : and, or, not
  * Data transfer/movement instructions
* Complete
  * It should be able to implement all the programs that users may write.

Features of an ISA – II

* Concise
  * The instruction set should have a limited size. Typically an ISA contains 32-1000 instructions.
* Generic
  * Instructions should not be too specialized, e.g. add14 (adds a number with 14) instruction is too specialized
* Simple
  * Should not be very complicated.

Designing an ISA

* Important questions that need to be answered :
  * How many instructions should we have ?
  * What should they do ?
  * How complicated should they be ?

Two different paradigms : RISC and CISC

RISC
(Reduced Instruction Set Computer)

CISC
(Complex Instruction Set Computer)

RISC vs CISC

A reduced instruction set computer (RISC) implements simple instructions that have a simple and regular structure. The number of instructions is typically a small number (64 to 128). Examples: ARM, IBM PowerPC, HP PA-RISC

A complex instruction set computer (CISC) implements complex instructions that are highly irregular, take multiple operands, and implement complex functionalities. Secondly, the number of instructions is large (typically 500+). Examples: Intel x86, AMD, VAX

CISC

Who developed it?
CISC emerged from **early computer architects in industry**, most notably at **IBM** and later **Intel**.

Key contributors:

- IBM engineering teams
- Intel processor architects

When?

- **1960s–1970s**

Why CISC?

- Early computers had **very limited memory**
- Goal: **reduce program size** by packing more work into a single instruction
- Instructions were often **complex, multi-cycle, and variable-length**

RISC

Who developed it?

RISC was pioneered in **academic and industrial research labs,** led by:

- John Cocke at IBM
- David Patterson at University of California, Berkeley
- John L. Hennessy at Stanford University

When?

Late 1970s – early 1980s

Why RISC?

- Empirical studies showed most programs use **simple instructions**
- Simpler instructions allow:
  - Pipelining
  - Higher clock speeds
  - Better compiler optimization

Feature	CISC	RISC
Instruction Set	Large and complex	Small and simple
Instruction Length	Variable	Fixed
Execution Time	One instruction may take multiple clock cycles	Most instructions execute in one clock cycle
Number of Instructions	Fewer instructions needed for a task	More instructions needed for the same task
Hardware Design	Complex	Simple
Control Unit	Microprogrammed	Hardwired
Memory Usage	Efficient (fewer instructions)	Less efficient (more instructions)
Power Consumption	Higher	Lower
Performance	Slower due to complex instruction decoding	Faster due to simple instructions and pipelining
Examples	Intel x86, AMD x86	ARM processors, Apple M-series, RISC-V

Summary Uptil Now ...

* Computers are dumb yet ultra-fast machines.
* Instructions are basic rudimentary commands used to communicate with the processor. A computer can execute billions of instructions per second.
* The compiler transforms a user program written in a high level language such as C to a program consisting of basic machine instructions.
* The instruction set architecture(ISA) refers to the semantics of all the instructions supported by a processor.
* The instruction set needs to be complete. It is desirable if it is also concise, generic, and simple.

Completeness of an ISA

How can we ensure that an ISA is complete ?

* Complete means :
  * Can implement all types of programs
  * For example, if we just have add instructions, we cannot subtract (NOT Complete)

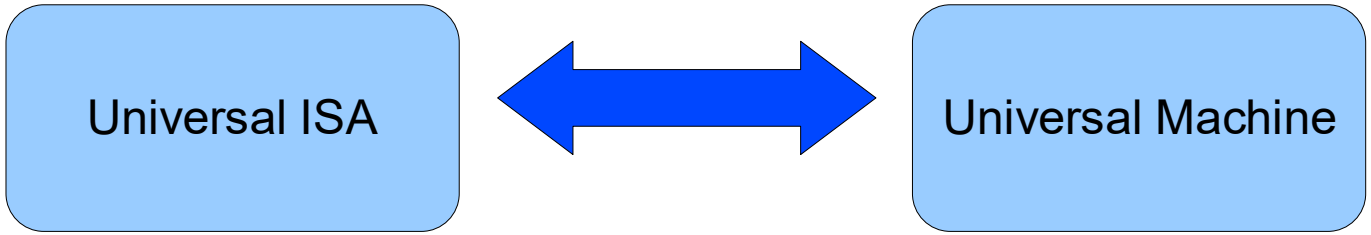
Completeness of an ISA – II

How to ensure that we have just enough instructions such that we can implement every possible program that we might want to write ?

Skip this part

Answer

* Let us look at results in theoretical computer science
* Is there an universal ISA ?

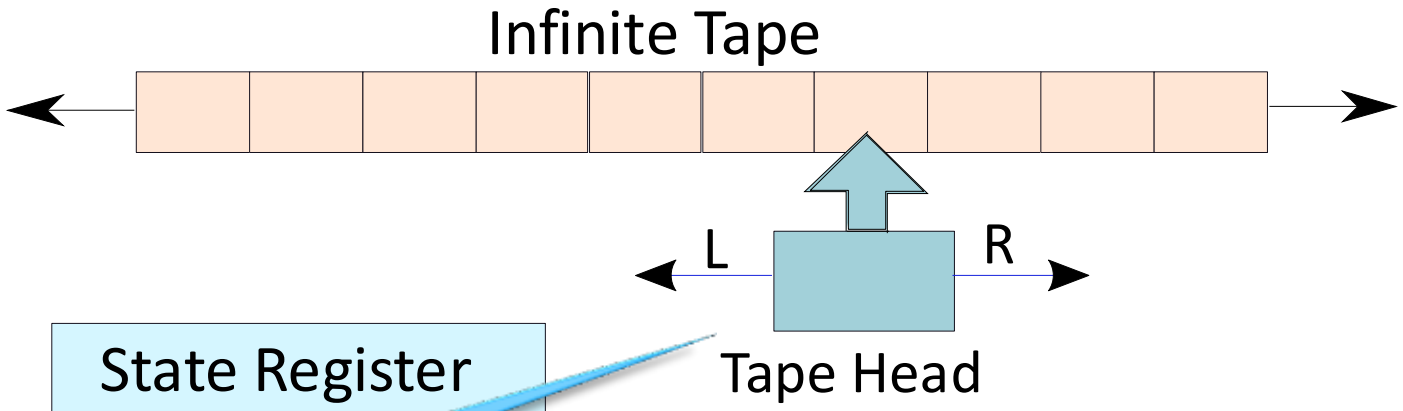


The universal machine has a set of basic actions, and each such **action** can be interpreted as an instruction.

The Turing Machine – Alan Turing

* Facts about Alan Turing
  * Known as the father of computer science
  * Discovered the Turing machine that is the most powerful computing device known to man
  * Indian connection : His father worked with the Indian Civil Service at the time he was born. He was posted in Chhatrapur, Odisha.

Turing Machine



State Register

The tape head can only move left or right

Action Table

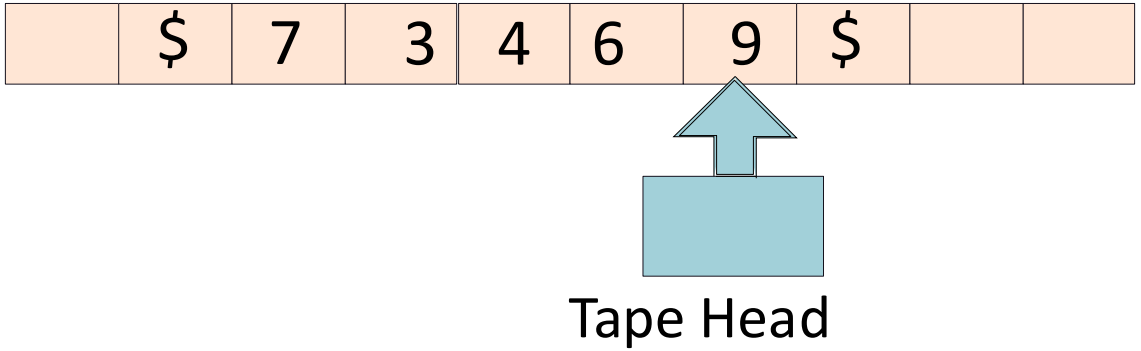
(old state, old symbol) -> (new state, new symbol, left/right)

Operation of a Turing Machine

* There is an inifinite tape that extends to the left and right. It consists of an infinite number of cells.
* The tape head points to a cell, and can either move 1 cell to the left or right
* Based on the symbol in the cell, and its current state, the Turing machine computes the transition :
  * Computes the next state
  * Overwrites the symbol in the cell (or keeps it the same)
  * Moves to the left or right by 1 cell
* The action table records the rules for the transitions.

Example of a Turing Machine

Design a Turing machine to increment a number by 1.



* Start from the rightmost position. (state = 1)
* If (state = 1), replace a number x, by x+1 mod 10
* The new state is equal to the value of the carry
* Keep going left till the '$' sign

More about the Turing Machine

* This machine is extremely simple, and extremely powerful
  * We can solve all kinds of problems – mathematical problems, engineering analyses, protein folding, computer games, …
  * Try to use the Turing machine to solve many more types of problems (TODO)

Church-Turing Thesis

Church-Turing thesis: Any real-world computation can be translated into an equivalent computation involving a Turing machine.
(source: Wolfram Mathworld)

* Note : It is a thesis, not a theorem
* For the last 60 years, nobody has found a counter-example

Definition:

Any computing system that is equivalent to a Turing machine is said to be Turing complete.

Universal Turing Machine

* For every problem in the world, we can design a Turing Machine (Church-Turing thesis)
* Can we design a universal Turing machine that can simulate any Turing machine. This will make it a universal machine (UTM)
* Why not? The logic of a Turing machine is really simple. We need to move the tape head left, or right, and update the symbol and state based on the action table. A UTM can easily do this.
* A UTM needs to have an action table, state register, and tape that can simulate any arbitrary Turing machine.

Universal Turing Machine